

Ex 3

Aim To perform data cleaning and data transformation on the dataset as part of Exploratory Data Analysis (EDA) using data science frameworks like CRISP-DM, SEMMA, and KDD.

Procedure

- **Data Cleaning:** Handled missing values (`dropna()`), removed duplicate records (`drop_duplicates()`), and cleaned column formatting by stripping whitespace and standardizing string casing.
- **Data Transformation:** Converted data types, restructured features for optimal storage, and transformed raw transaction records into structured data objects.
- **Pipeline Integration:** Exported cleaned and transformed data into CSV and JSON files, and migrated the records into an SQLite database for querying.

Program:

```
import pandas as pd
df = pd.read_csv("../content/001/retail1.csv", encoding="latin1")

df.head()

display(df.describe())

print(df.shape)

df.info()

(541360, 8)
<class 'pandas.core.frame.DataFrame'>
Int64Index: 541360 entries, 0 to 541359
Data columns (total 8 columns):
 #   column            non-null count  type
---  ---
 0   InvoiceNo         541360 non-null object
 1   StockCode        541360 non-null object
 2   Description      540455 non-null object
 3   Quantity         541360 non-null int64
 4   InvoiceDate       541360 non-null object
 5   UnitPrice        541360 non-null float64
 6   CustomerID       493218 non-null float64
 7   Country          541360 non-null object
dtypes: float64(2), int64(1), object(5)
memory usage: 33.1+ MB
```

```
print("Missing Values")

print(df.isnull().sum())

[6]

-- Missing Values

InvoiceNo      0
StockCode      0
Description    1454
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID    135080
Country        0
dtype: int64

D print(df.isnull().sum())

[7]

-- InvoiceNo      0
StockCode      0
Description    1454
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID    135080
Country        0
dtype: int64

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```

```
print("Duplicate Rows :", df.duplicated().sum())

df.drop_duplicates(inplace=True)

print("After removal :", df.shape)

Duplicate Rows : 5268
After removal : (52660, 8)

Q1 = df["Quantity"].quantile(0.25)
Q3 = df["Quantity"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers_quantity = df[
    (df["Quantity"] < lower) |
    (df["Quantity"] > upper)
]

print("Outliers in Quantity column")
display(outliers_quantity.head())

Outliers in Quantity column
```

```
D df = df[df["Quantity"] > 0]
df = df[df["UnitPrice"] > 0]

print("Dataframe shape after removing non-positive quantity and unitprice:", df.shape)
print("Summary statistics after cleaning:")
display(df.describe())

[8]

-- Dataframe shape after removing non-positive quantity and unitprice: (524870, 8)
Summary statistics after cleaning:

% Generate + Code + Markdown

print(df.info())

[9]

-- class 'pandas.core.frame.DataFrame'
Index: 524870 entries, 0 to 524869
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   InvoiceNo    524870 non-null  object
 1   StockCode   524870 non-null  object
 2   Description  524870 non-null  object
 3   Quantity    524870 non-null  int64
 4   InvoiceDate  524870 non-null  object
 5   UnitPrice   524870 non-null  float64
 6   CustomerID  306602 non-null  float64
 7   Country     524870 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 35.8+ MB
None
```

```
print("Missing values after initial cleaning:\n", df.isnull().sum())
```

Python

Missing values after initial cleaning:

```
InvoiceNo      0
StockCode      0
Description    0
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID     0
Country        0
dtype: int64
```

```
from sklearn.preprocessing import MinMaxScaler, LabelEncoder

numeric_cols_retail = ['Quantity', 'UnitPrice']
categorical_cols_retail = ['Country']

scaler = MinMaxScaler()
df[numeric_cols_retail] = scaler.fit_transform(df[numeric_cols_retail])

print("Scaled numeric columns (Quantity, UnitPrice):")
display(df[numeric_cols_retail].head())
```

Python

Scaled numeric columns (Quantity, UnitPrice):

```
df.to_csv("cleaned_online_retail_dataset.csv", index=False)
print("Cleaned Online Retail dataset saved successfully as 'cleaned_online_retail_dataset.csv'.")
```

[20]

Python

— Cleaned Online Retail dataset saved successfully as 'cleaned_online_retail_dataset.csv'.

```
df.to_csv("cleaned_online_retail_dataset.csv", index=False)
print("Cleaned Online Retail dataset saved successfully as 'cleaned_online_retail_dataset.csv'.")
```

[20]

Python

— Cleaned Online Retail dataset saved successfully as 'cleaned_online_retail_dataset.csv'.

```
df.dropna(subset=['CustomerID'], inplace=True)
print("Shape after dropping rows with missing CustomerID:", df.shape)
```

[20]

Python

— Shape after dropping rows with missing CustomerID: (392692, 8)

```
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
print("InvoiceDate column converted to datetime.")
```

[20]

Python

— InvoiceDate column converted to datetime.

```
print(df.info())
print("Missing values after cleaning and type conversion:\n", df.isnull().sum())
```

[20]

Python

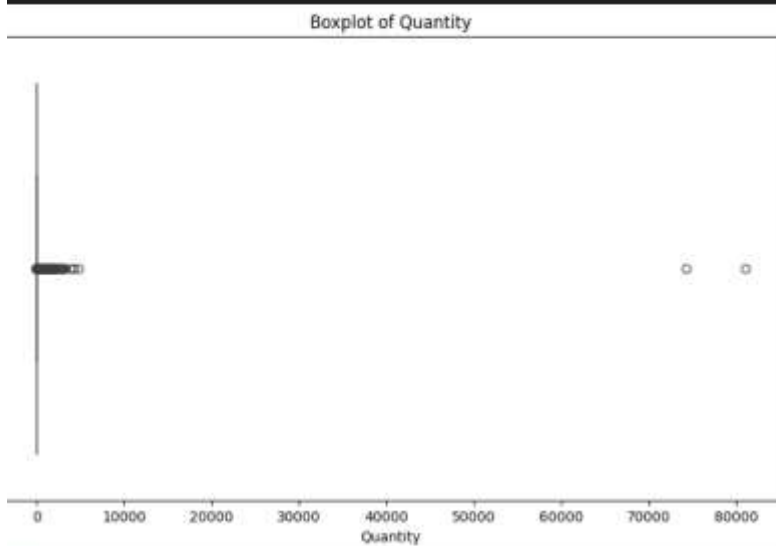
```
<class 'pandas.core.frame.DataFrame'>
Index: 392692 entries, 0 to 343988
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ---
0   InvoiceNo    392692 non-null  object
1   StockCode   392692 non-null  object
2   Description  392692 non-null  object
3   Quantity    392692 non-null  float64
4   InvoiceDate  392692 non-null  datetime64[ns]
5   UnitPrice   392692 non-null  float64
6   CustomerID  392692 non-null  float64
7   Country     392692 non-null  int64
```

```
print(df.info())
print("Missing values after cleaning and type conversion:\n", df.isnull().sum())
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 392692 entries, 0 to 343988
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ---
0   InvoiceNo    392692 non-null  object
1   StockCode   392692 non-null  object
2   Description  392692 non-null  object
3   Quantity    392692 non-null  float64
4   InvoiceDate  392692 non-null  datetime64[ns]
5   UnitPrice   392692 non-null  float64
6   CustomerID  392692 non-null  float64
7   Country     392692 non-null  int64
dtypes: datetime64[ns](1), float64(3), int64(1), object(3)
memory usage: 27.6+ MB
None
Missing values after cleaning and type conversion:
InvoiceNo      0
StockCode      0
Description    0
Quantity       0
InvoiceDate    0
UnitPrice      0
CustomerID     0
Country        0
dtype: int64
```

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))
sns.boxplot(x=df['Quantity'])
plt.title('Boxplot of Quantity')
plt.xlabel('Quantity')
plt.show()
```



Result Thus the EDA, data cleaning, and data transformation using data science frameworks like CRISP-DM, SEMMA, KDD is implemented successfully.